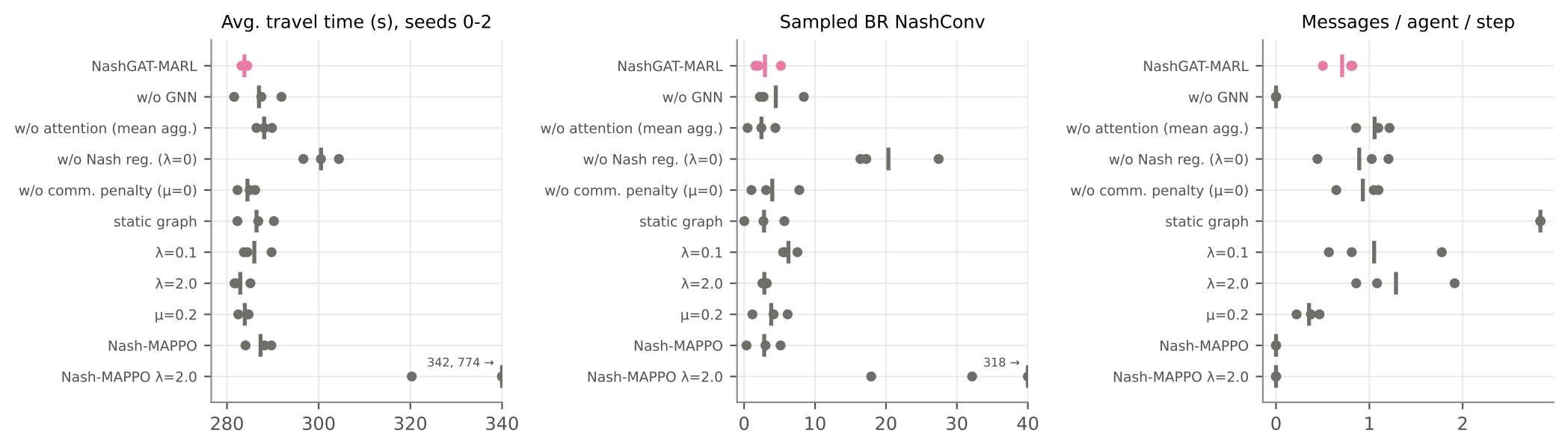
Avg. travel time (s), seeds 0-2
NashGAT-MARL
w/o GNN
w/o attention (mean agg.)
w/o Nash reg. (λ=0)
w/o comm. penalty (μ=0)
static graph
λ=0.1
λ=2.0
μ=0.2
Nash-MAPPO
Nash-MAPPO λ=2.0
342, 774 →
280
300
320
340
Sampled BR NashConv
NashGAT-MARL
w/o GNN
w/o attention (mean agg.)
w/o Nash reg. (λ=0)
w/o comm. penalty (μ=0)
static graph
λ=0.1
λ=2.0
μ=0.2
Nash-MAPPO
Nash-MAPPO λ=2.0
318 →
0
10
20
30
40
Messages / agent / step
NashGAT-MARL
w/o GNN
w/o attention (mean agg.)
w/o Nash reg. (λ=0)
w/o comm. penalty (μ=0)
static graph
λ=0.1
λ=2.0
μ=0.2
Nash-MAPPO
Nash-MAPPO λ=2.0
0
1
2